Background-Preserving Pedestrian Insertion with Stable Diffusion 3.5: A Rule-Guided Pipeline and LoRA Study

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Abstract

This report presents a background-preserving pedestrian augmentation system based on Stable Diffusion 3.5. Two flows are studied: the Rule-Guided SD3.5 Pipeline, used as the primary controlled-augmentation flow, and a concept LoRA segment-and-paste flow, used as a secondary appearance-generation baseline. Rule-Guided uses target-scene context and applies placement control, expected-height scale correction, grounding and occlusion checks, mask and edge filtering, quality gates, and bounded retry before accepting an output. Across three retained smoke runs, it accepted 103 of 150 generated candidates, giving a weighted acceptance rate of 0.687 and an unweighted mean scale score of 0.957. The rejection log identified invalid mask geometry and poor placement as the main recorded failure causes, accounting for 28 and 11 of 47 rejected candidates, respectively. The LoRA flow achieved favourable raw detector-count and background-preservation values, but its current implementation does not use target-scene placement awareness, expected-height scale correction, grounding validation, or a dedicated edge-quality rejection stage. The two available shared-metric exports use different source cases and effective-support-mask conventions; they are therefore descriptive only and are not a direct benchmark. Under current evidence, Rule-Guided is the more production-ready controlled-augmentation flow because it provides the scene-aware acceptance, rejection, and tuning controls required for dataset production. A future paired evaluation on common frozen scenes will add target-specific detection-overlap metrics, valid scale error, and blinded manual review before any cross-flow quality ranking is claimed.

1 Introduction

Global context and motivation

Supervised object detectors and segmenters remain data-hungry, and the classes that matter most are often the least represented: rare poses, crowded scenes, unusual scales. Synthetic object insertion - placing a new, plausible instance into a real image - is an attractive way to augment such data, because it promises new positives without new photography or new manual labels. The promise holds only if the inserted object is realistic enough to be a valid positive and all pixels outside a clearly defined composite support are preserved under a documented image-processing convention, so that every annotation already attached to the scene remains interpretable.

Figure 1: One source image, one inserted pedestrian, and the source-to-composite difference map for the Rule-Guided SD3.5 Pipeline segment-and-composite flow (CityPersons `aachen_000139`, `add_occluded_pedestrian`). The difference (right; blue = 0, warmer = larger) is concentrated near the inserted-object support: outside the annotated support the mean per-pixel RGB difference is 0.4/255, while the object reaches 171/255. This is consistent with source-image compositing, but it is not by itself a bit-exact preservation proof because the measured region depends on the effective mask and image processing used in the export.

I use CityPersons pedestrian insertion [1] as the reference experiment. Pedestrians are a deliberately hard, useful test case: they span a wide range of scales, must be grounded on a plausible surface (feet on the ground, not floating), are frequently occluded by scene structure, and admit a strong off-the-shelf validator in a person detector [2]. A method that handles pedestrians well therefore exercises scale, grounding, occlusion, and detector-based validation at once. The task is nonetheless framed as object insertion in general, not pedestrians specifically; the detector, prompt set, placement policy, and validation rules are the parts that specialise to a class.

The cost of getting it wrong is imbalance. A convincing pedestrian pasted onto a background whose pixels have shifted is worse than no augmentation at all: the new instance may be a usable positive, but every pre-existing box, mask, or class label on that frame is now attached to pixels that no longer match it. Background corruption silently poisons the dataset, which motivates an architecture in which background fidelity is not something the generator is merely asked to preserve but something the pipeline constrains through source-image compositing.

Literature review and research gaps

Insertion and editing methods can be read as successive attempts to enlarge the canvas the model may touch while shrinking the damage it does to everything else; each generation removes one failure mode and exposes the next.

- Full-image img2img - adds objects but rewrites the global scene.
- Patch / tight-mask / bbox inpaint - progressively more canvas, but generated background inside the edit region still leaks in.
- Training-free attention-injection insertion (e.g. Add-it [3]) - strong placement and no fine-tuning, but it blends in latent space and decodes the whole frame, so pixels outside the object are not guaranteed identical to the source, and its constants are derived for a different (FLUX) backbone.

These limitations share a root cause: background preservation is treated as a behaviour to be encouraged (by a mask, a prompt, or an attention bias) rather than a property to be enforced.

The gap addressed here is therefore twofold: an architecture that makes preservation structural across heterogeneous flows, and a single metric core that can support a paired comparison once matched evidence is exported for each flow.

Problem statement

Let $I_{\text{src}} \in \mathbb{R}^{H\times W\times 3}$ denote a source image and let c be a target-object specification (class, prompt, optional placement prior). Let $M_{\text{eff}} \in \{0,1\}^{H\times W}$ denote the effective composite support: the final region that may be altered after mask erosion, feathering, colour transfer, shadowing, and any blend operation. The insertion task is to learn or apply a map $\mathcal{F} : (I_{\text{src}}, c) \to I_{\text{out}}$, with $I_{\text{out}} \in \mathbb{R}^{H\times W\times 3}$, satisfying four requirements: (i) object validity - a detectable, correctly-classed object of plausible appearance is added; (ii) scale and grounding - the object's size and contact point are physically plausible for the scene; (iii) background preservation - pixels outside M_{eff} should remain as close as possible to I_{src} under the exported-image convention (BG_MAE $\to 0$, BG_SSIM $\to 1$); and (iv) label integrity - the augmented sample remains valid training data for the downstream task.

Three challenges constrain any solution. (1) Diffusion denoising can alter any visible pixel. A single pass that re-noises and re-decodes the frame has no mechanism that leaves a chosen region unchanged without an explicit restoration or composite operation, so background preservation must be enforced architecturally (segment the object and composite onto the original), not requested through a prompt or a mask alone. (2) Plausible appearance does not imply correct scale. Generated objects routinely look realistic yet sit at the wrong physical size or float off the ground, because the generator has no explicit geometric prior; scale and grounding must be corrected after generation from depth or placement cues. (3) Detector-guided validation is practical but not human-quality. Using a detector as both extractor and validator makes the pipeline self-checking and reproducible, but detector confidence is a proxy: it can accept an artefact a human would reject, so quantitative gains must be read alongside qualitative inspection.

Contributions

The principal contributions of this report are:

1. A background-as-source-of-truth insertion architecture: The source image is the trusted scene and generated background is discarded: only the retained object support is composited back. Preservation is therefore implemented as a pipeline constraint rather than a prompt-level preference.

2. A rule-guided primary augmentation pipeline: Rule-Guided SD3.5 Pipeline combines context-conditioned generation with placement policy, expected-height scale correction, ground-contact checks, occlusion handling, mask and edge filtering, validation, and bounded retry. The saved diagnostics retain 103/150 accepted candidates and a mean flow-specific scale score of 0.957 across three named smoke runs.

3. A transparent audit of the secondary concept-LoRA flow: The supplied $N = 100$ LoRA shared-metric CSV is internally consistent for row identity and detector-count arithmetic, but it has no usable scale result and does not implement target-scene placement awareness or a dedicated edge-quality filtering gate. These limitations are treated as future-work requirements, not hidden by raw pixel metrics.

4. A task-readiness decision rule: The report distinguishes descriptive raw shared metrics from the controls required for safe dataset production. Rule-Guided SD3.5 Pipeline is selected as the primary pipeline because it has the scene-aware acceptance and rejection

mechanisms required by controlled pedestrian augmentation; LoRA is retained as a secondary appearance-generation baseline.

2 Objectives

General objective

The overall objective is to establish a background-preserving pedestrian-insertion workflow that can produce controlled synthetic training examples. Rule-Guided SD3.5 Pipeline is the principal method because it observes local scene context and rejects candidates that fail placement, scale, grounding, occlusion, mask-quality, or detection checks. The concept LoRA is evaluated as a secondary appearance-generation baseline whose missing scene controls are explicitly identified for future work.

Research questions

RQ1. Can source-image compositing constrain background change more reliably than asking a diffusion model to preserve the scene through prompting alone?

RQ2. Does Rule-Guided SD3.5 Pipeline provide the controls needed for safe and scene-plausible pedestrian augmentation: target-scene context, placement awareness, scale correction, grounding, occlusion handling, edge filtering, validation, and retry?

RQ3. What do the audited LoRA metrics establish about object addition and background change, and which controlled-augmentation properties remain unmeasured or unavailable because LoRA currently lacks scale correction, placement awareness, and dedicated edge filtering?

Specific objectives

1. Architecture. Build an insertion architecture in which generated background is discarded and the source image supplies the final canvas outside the effective composite support M_{eff}.

2. Primary-pipeline control. Implement scene-conditioned placement, scale correction, grounding, occlusion, mask and edge filtering, validation, and retry in Rule-Guided SD3.5 Pipeline. Criterion: report accepted / generated candidates, per-run acceptance, flow-specific scale score, and rejection reasons for all retained smoke runs.

3. LoRA metric integrity. Audit the exported LoRA CSV before using it as evidence. Criterion: verify row identity, seed uniqueness, detector-count arithmetic, missingness, and the consistency of OBJECT_ADDED with positive count delta.

4. Method selection. Select the main dataset-production flow using the controls required by controlled augmentation, not only raw pixel similarity. Criterion: the chosen flow must expose target-scene placement, scale and grounding checks, occlusion handling, edge-quality filtering, acceptance logging, and bounded retry.

5. Future LoRA improvements. Define the missing controls that must be added before LoRA can serve as a controlled-augmentation flow: placement awareness, expected-height scale correction, grounding and occlusion validation, dedicated edge filtering, and retry.

Table 1: Generation backbones considered, and why each is not the default, following `docs/model-selection.md`.

Model	Why not the default
SD3 Medium	Weaker prompt adherence than SD3.5.
SD3.5 Medium NF4	Low-VRAM fallback; full precision mode must be logged per run.
SD3.5 Large Turbo	Heavier; less ideal for a retry-heavy pipeline.
FLUX.1-dev FP16	Strong quality, impractical VRAM/runtime here.

3 Materials and Methods

3.1 Model selection

Both flows use Stable Diffusion 3.5 Medium as their generation backbone [4]. The backbone is a candidate generator only: a detector/segmenter extracts the proposed object and the compositing stage returns it to the source image. Selection was driven less by peak photorealism than by suitability for a retry-heavy, segmentation-and-composite pipeline: object realism, prompt adherence, controllable img2img behaviour, stability under retries, practicality on a Kaggle T4-class 16 GB environment, and compatibility with detector-guided validation. The project configuration considers FP16/BF16 and keeps 4-bit NF4 as a low-VRAM fallback. Because the saved archive does not retain an immutable per-run dtype log, this report treats the exact precision used for the reported metrics as a configuration item to be verified from run provenance rather than as a separately measured result. Table 1 lists the backbones considered and why each is not the default.

3.2 Diffusion backbone: MM-DiT and rectified flow

Both flows share a single generation backbone - Stable Diffusion 3.5 Medium - so their architectural differences lie entirely in how each conditions that backbone, not in the backbone itself. I summarise the backbone here so the per-flow descriptions can refer to it.

From U-Net to a latent transformer

Like prior latent diffusion models [5], SD3.5 operates not on pixels but in the latent space of a pretrained autoencoder: an image $I \in \mathbb{R}^{H\times W\times 3}$ is encoded to a latent $x \in \mathbb{R}^{h\times w\times c}$ (here $h = w = 64$, $c = 16$ for a 512 px image), denoised there, and decoded back. The denoiser is a Diffusion Transformer (DiT) [6] rather than a U-Net: the latent is split into non-overlapping patches, each linearly embedded into a token, and the resulting token sequence is processed by a stack of transformer blocks. DiT injects the conditioning signal (timestep, class) through adaptive layer normalisation (adaLN-Zero) - per-block scale/shift parameters regressed from the conditioning embedding - which [6] found to outperform both cross-attention and in-context token conditioning.

MM-DiT: two streams, one attention

SD3 generalises DiT to text-to-image with the multimodal MM-DiT block [7]. Image patches (with positional encodings, from 2×2 latent patches) and text tokens are embedded to a common width and concatenated into one sequence; two separate sets of weights process the two modalities - effectively two parallel transformers - that are joined only for the attention operation, so each modality keeps its own representation while still attending to the other [7]. Text conditioning comes from three frozen encoders (CLIP-L, CLIP-G, T5-XXL): the pooled CLIP vector plus the timestep drive the adaLN modulation, while the T5/CLIP token sequence enters as the text stream. In our runs T5-XXL is disabled by default to fit the T4 16 GB budget, leaving the two CLIP encoders; this trades some fine-grained prompt fidelity for memory, which is acceptable here because the prompts are short object descriptions rather than long compositional scenes.

Rectified flow

SD3.5 is trained as a rectified-flow model [7]: rather than the DDPM variance schedule, the forward process is the straight-line interpolation

$$X_t = (1 - \sigma_t)\, x_0 + \sigma_t\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),\ \ \sigma_t \in [0, 1], \tag{1}$$

and the network predicts the velocity v_θ along this path. Sampling integrates the corresponding ODE with a FlowMatch-Euler scheduler. Figure 2 shows the generation flow common to both flows.

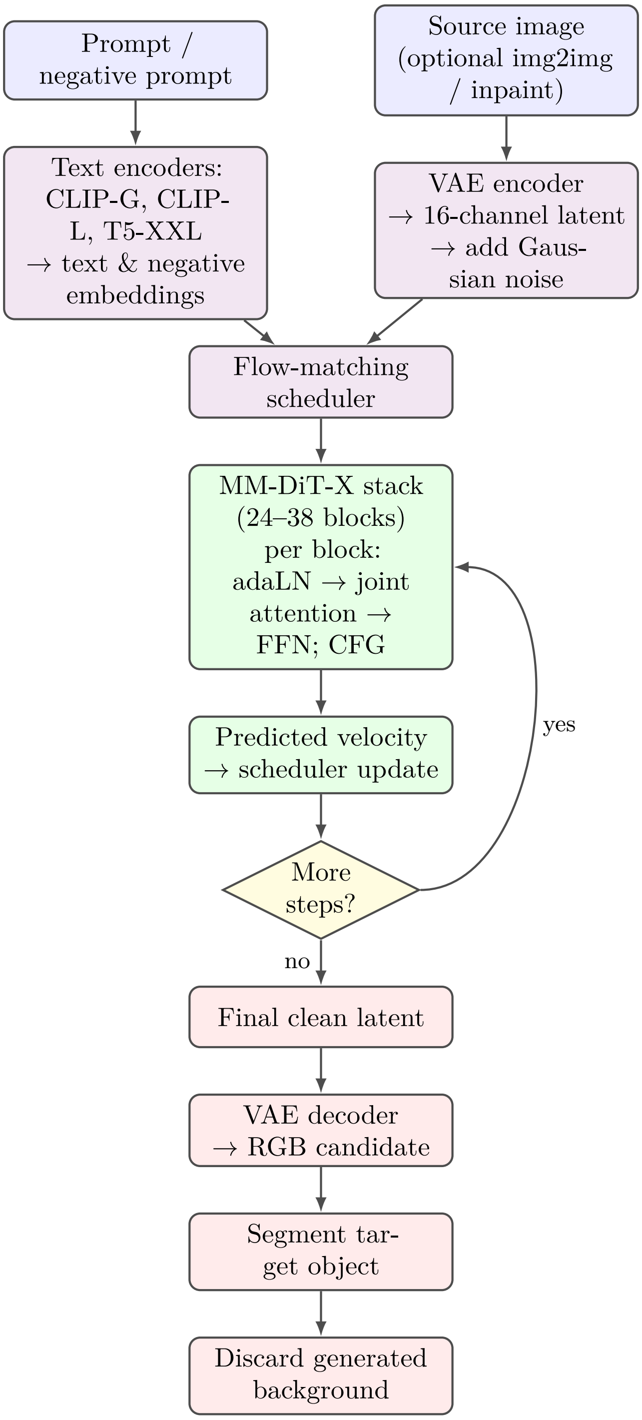

Figure 2: The SD3.5 generation flow used inside the context/candidate generation step of every flow (re-drawn from `docs/diagram.md`). Text conditioning (the three encoders; T5-XXL disabled by default here) and an optionally img2img-encoded latent feed the flow-matching denoise loop over the MM-DiT-X stack; the final latent is decoded, the target object is segmented, and the generated background is discarded - the point at which background-as-source-of-truth begins.

Table 2: Dataset roles separated by what is defined in the project protocol versus what is retained in the saved report archive. "Not retained" means the report cannot independently substantiate the corresponding run-level contribution.

Source	Role in protocol	Evidence retained for this report
CityPersons [1]	Pedestrian reference domain; source backgrounds for rule-guided and qualitative composite examples; valid/test intended frozen benchmark.	Frankfurt/Aachen examples are present; per-row evaluation provenance is not retained.
MOT17-02	Candidate ETL source; group by sequence-time window.	Available in the data design; no post-filter contribution log retained.
Human detection	Candidate positive/background source; group by perceptual cluster.	Available in the data design; no post-filter contribution log retained.
PIPE [9]	Person-class generation targets for the stated concept-LoRA configuration.	Referenced by the configuration narrative; exact retained training count and release manifest are absent.

3.3 Dataset and split protocol

The repository contains a broader data-engineering design than the saved experiment archive. To avoid conflating a candidate source with data proven to have been used in the reported adapter, this section distinguishes three roles: (1) sources available to the ETL or rule-guided pipeline, (2) the concept-training source stated for the saved LoRA configuration, and (3) source backgrounds visible in the saved qualitative exports.

The reported concept-LoRA configuration identifies the person-class subset of Paint-by-Inpaint / PIPE [9] as the generation-target source. PIPE contains 888,000 train and 752 test items at 512×512 across 824 `Instruction_Class` categories. The loader is intended to retain only whole-word person classes; the before/after pairing is not required for the plain text-to-image concept objective. CityPersons [1] supplies the pedestrian reference domain and is visibly used for the saved Frankfurt/Aachen composites. MOT17-02 [8] and the human-detection collection are candidate sources in the project data contract, but the saved report archive does not contain a release manifest that proves their post-filter contribution to the LoRA artifact.

The intended split policy is group-aware: CityPersons by scene, MOT17 by sequence window, and the human-detection collection by perceptual-duplicate cluster. The CityPersons validation/test material should remain frozen for downstream evaluation and must not be used for LoRA training, prompt tuning, threshold tuning, or synthetic background selection. The current archive does not retain a complete release manifest, post-filter counts, or the background-source provenance for every one of the 100 LoRA metric rows. Accordingly, this report does not claim that those 100 rows constitute a PIPE test-set evaluation or a fully verified CityPersons benchmark.

3.4 Future paired smoke-test protocol

A future controlled smoke test should use four source sets with ten source images per set, giving 40 source scenes. Each scene can receive up to five candidate attempts, yielding a planned budget of

$$4 \text{ source sets} \times 10 \text{ images per set} \times 5 \text{ attempts per image} = 200 \text{ candidate attempts.} \quad (2)$$

This 200-candidate budget is a future validation design and is not reported as an observed sample size in this study. The retained archive instead provides three Rule-Guided summaries over 150 candidates, a Rule-Guided shared-metric export with 32 rows, and a LoRA shared-metric export with 100 rows.

For a valid paired comparison, the evaluation set must be split before tuning. A small development subset may tune thresholds and image-processing controls, while the final 40 source scenes remain frozen. Both flows should use the same source image, target prompt, seed where supported, detector version and threshold, output resolution, and effective-support mask convention. This protocol prevents different test conditions from being interpreted as a performance gain.

3.5 Flow A - Rule-Guided SD3.5 Pipeline (Primary pipeline)

This report designates Rule-Guided SD3.5 Pipeline as the primary augmentation flow. It treats the MM-DiT backbone (Section 3.2) as a black-box img2img candidate generator and places all background-preservation logic outside it. The motivating observation is architectural rather than empirical: because rectified-flow denoising (Equation (1)) can in principle move any latent token, no prompt or guidance setting guarantees that pixels outside the intended object stay fixed - so Rule-Guided SD3.5 Pipeline never asks the model to preserve them. Instead it generates a candidate, segments only the new object, discards the generated background, and composites the object onto the untouched source. The original image is the source of truth; generated background pixels are never used as the final canvas outside the effective composite support. The pipeline is a fixed sequence of explicit stages:

$$\begin{aligned} &\text{placement} \rightarrow \text{context generation} \rightarrow \text{segmentation} \rightarrow \\ &\text{scale correction} \rightarrow \text{compositing} \rightarrow \text{validation} \rightarrow \text{retry.} \end{aligned} \quad (3)$$

Stages

(1) Placement proposes a valid insertion region from object-specific priors (road or sidewalk masks, existing person/vehicle boxes, depth ordering, overlap checks) - placement is a policy, not a prompt, since a realistic object in an invalid location still corrupts the dataset. (2) Context generation runs SD3.5 img2img over a local context crop, so the model has enough surrounding scene to produce a coherent, correctly-lit candidate (the lesson from the V1–V4 ablations: too little canvas yields ghosts; too much rewrites the scene). (3) Segmentation extracts the new object with YOLOv8m-seg. (4) Scale correction estimates expected pixel height from depth/ground-position, resizes the crop and mask together with the contact point (feet) anchored, and rejects objects outside a plausible size envelope - geometry, not text, governs scale. (5) Compositing cleans the mask, trims the generated-background fringe, applies occlusion rules against foreground persons/vehicles, harmonises colour/brightness/texture/edge (optional contact shadow), and alpha-pastes only the object pixels. (6) Validation + retry re-runs the detector and geometry/background checks; weak samples are rejected and regenerated under a bounded budget rather than saved.

Table 3: Rule-guided tuning functions: Each function changes a defined part of the pipeline and is linked to a recorded failure type. Exact values must be saved in the run configuration.

Function	Controls	Used when	Output
Placement tuning	Safe-region score, depth range, object clearance, foot-anchor location.	Bad placement or bad depth overlap.	Valid placement proposal or rejection.
Context-generation tuning	Context crop margin, denoise strength, guidance scale, inference steps.	Candidate has poor local lighting, shape, or scene fit.	New candidate image.
Mask tuning	Detection threshold, mask erosion, feather radius, small-component removal.	Mask aspect is invalid or the boundary has a halo.	Clean object mask.
Geometry tuning	Expected-height envelope, scale correction, ground-contact tolerance.	Floating person or unrecoverable scale.	Resized crop and anchored mask, or rejection.
Blend tuning	Colour matching and edge blending.	Visible seam or sticker effect.	Lower seam risk.
Acceptance tuning	Quality-score threshold τ_Q, component thresholds, maximum retries.	Candidate fails the final safety gate.	Accept, record reason, or retry.

Quality score and safety gate

Rule-Guided SD3.5 Pipeline keeps a fused quality score as a flow-specific extension (not part of the shared core of Section 3.8):

$$Q = 0.45\, s_{\text{obj}} + 0.25\, s_{\text{scale}} + 0.20\, s_{\text{bg}} + 0.10\, s_{\text{edge}}, \tag{4}$$

used for analysis and conservative autotune guidance, not as a claim of absolute image quality. By policy augmentation is restricted to the `train` split so it cannot contaminate the evaluation benchmark - `target_splits` is set to `train` only. Figure 3 draws the staged flow of Equation (3), with the trusted-background path highlighted.

Tuning functions and safe parameter updates

The Rule-Guided SD3.5 Pipeline exposes tuning controls so that a failed candidate can be diagnosed and improved without silently weakening the evaluation protocol. Tuning is performed only on a development subset. Once the final smoke set is frozen, parameters are fixed and no longer changed from one test image to the next.

The flow-specific quality score in Equation (4) is used to rank or reject candidates after the required hard checks. A safe tuning rule is

$$\begin{gathered} \theta^* = \arg\max_{\theta \in \Theta} \frac{1}{|D_{\text{dev}}|} \sum_{x \in D_{\text{dev}}} Q(x; \theta), \\ \text{subject to a fixed benchmark split, detector, and acceptance policy on } D_{\text{test}}. \end{gathered} \tag{5}$$

Here θ contains the tunable controls in Table 3. The constraint is important: it prevents the final smoke set from becoming a tuning set.

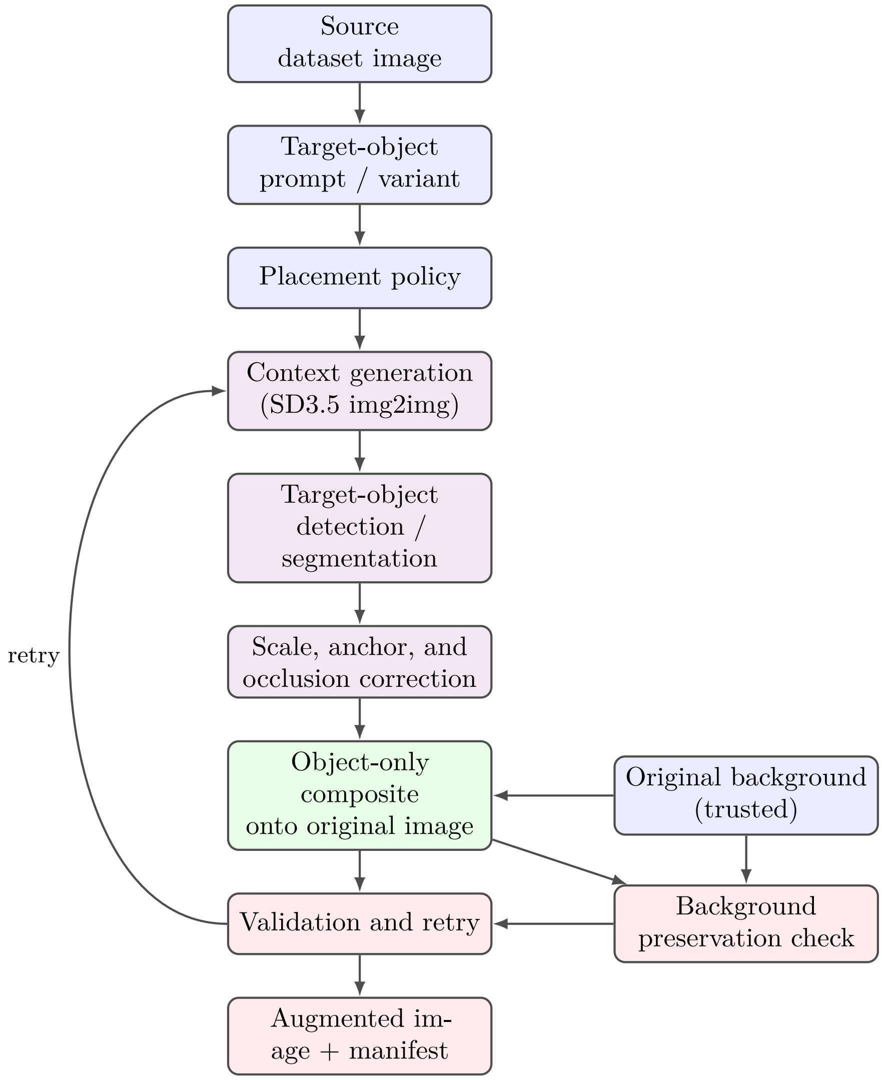

Figure 3: The Rule-Guided SD3.5 Pipeline context–object composite flow of Equation (3) (re-drawn from `docs/diagram.md`). The blue original background is the trusted source: it is composited under the segmented object and is the reference for the background-preservation check, so pixels outside the effective composite support are never taken from generation. Failed validations retry the generation stage under a bounded budget.

3.6 Flow B - Concept LoRA + segment-and-paste (Secondary baseline)

The concept LoRA is retained as a secondary appearance-generation baseline, not as the primary dataset-production path. It separates what to generate from which pixels to keep: a LoRA learns a pedestrian concept, a segmenter extracts the generated person, and the extracted person is pasted onto an original source image. The current implementation does not condition generation on the target scene before the person is created. Consequently, it does not currently provide target-scene placement awareness, expected-height scale correction, explicit ground-contact correction, or a dedicated edge-quality filtering and rejection stage.

Training: a plain text→image concept LoRA

A single LoRA is trained, DreamBooth-style, so the frozen SD3.5 backbone learns to generate images in the distribution of the PIPE "add a person" subset. There is no mask, no source image, and no image-conditioning projection: training is the standard SD3 rectified-flow objective on the target latent alone,

$$z_t = (1 - \sigma_t)\, z_{\text{img}} + \sigma_t\, \epsilon, \qquad \mathcal{L} = w(\sigma_t)\, \big\| \hat{v}_\theta(z_t, t, c_{\text{text}}) - (\epsilon - z_{\text{img}}) \big\|^2, \tag{6}$$

where $z_{\text{img}} = \text{VAE(image)}$ and c_{text} is the precomputed text embedding. Only the attention LoRA adapters are trained; the MM-DiT weights and text encoders stay frozen, and text embeddings are precomputed to disk so the encoders can be dropped during the T4-budget run. Inference therefore needs a single adapter artifact and no input projection.

Why PIPE images

The concept LoRA is trained on the person-class subset of PIPE [9]: real photographs that contain people, used as plain generation targets. The before/after pairing that PIPE was built for is not required for this concept objective, because the LoRA generates rather than edits a given scene. The loader filters strictly on the `Instruction_Class` field; the free-text caption is not used as a class selector. The exact retained person-image count is logged by the loader at training time.

Generate → segment → paste

Background preservation is an inference-time composite, not a training objective:

1. Generate. Base SD3.5 with the trained LoRA produces a full image containing a person from text alone. The model does not see the target source scene, so the generated person is not conditioned on the target scene's position, geometry, or lighting.
2. Segment. YOLOv8-seg returns a per-person mask, bounding box, and confidence from the generated frame; low-confidence or tiny detections can be dropped.
3. Paste. The segmented person is composited onto the original background. The current path may use basic mask erosion, feathering, and colour transfer to reduce seams. These are blend operations, not a dedicated edge-quality filtering and rejection system. The implementation does not currently calculate an expected target height, correct scale against scene geometry, or validate whether the final placement is physically safe.

The auxiliary LoRA flow can generate pedestrian appearance, but its person was created under a different generated scene and may not match the target scene's scale, viewpoint, ground plane, or lighting. This is the central reason it is not selected as the main controlled-augmentation flow. Placement awareness, scale correction, ground-contact validation, occlusion reasoning, dedicated edge filtering, and retry are listed as future LoRA upgrades in Section 4.5.

Table 4: Two image types produced by the study: The outputs are stored and interpreted separately because their generation inputs, controls, and validation rules are different.

Output type	**Generation input**	**Controls before acceptance**	**Role**
Rule-guided accepted composite	Source scene, local context crop, target prompt, placement proposal.	Placement, expected-height scale correction, foot anchor, occlusion, mask and edge filtering, quality gate, retry.	Primary controlled-augmentation output.
LoRA segment-and-paste composite	Text prompt for a person, then a separate source background for paste.	Basic segmentation and blending only; no target-scene placement awareness, scale correction, or dedicated edge filter.	Secondary appearance-generation baseline.

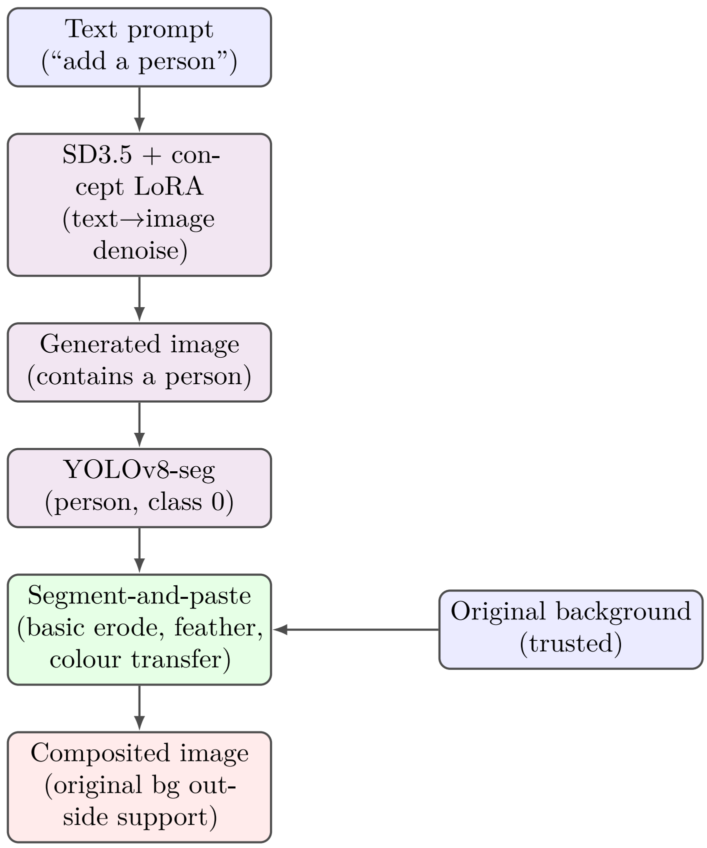

Figure 4: The concept-LoRA generate-then-paste flow. It generates a person from text, segments that person, and pastes the result onto a trusted source image. The flow does not currently include target-scene placement awareness, scale correction, or a dedicated edge-quality filtering and rejection stage.

3.7 Two flows with different output types

The two flows share a final compositing idea but produce different types of output. Rule-Guided SD3.5 Pipeline generates a candidate in the target scene context, corrects it against that scene, and retains an image only after safety checks pass. The LoRA baseline first generates a person without target-scene context and then pastes the segmented person onto a source image. The study keeps these outputs separate so that a controlled, accepted training image is not confused with an appearance-only baseline output.

The Rule-Guided flow observes local scene context before candidate generation and applies placement, depth, scale, grounding, occlusion, edge-quality, validation, and retry controls. The concept LoRA does not use the target scene to generate the person and therefore cannot currently guarantee that the output has the correct scale, safe placement, or clean edge in that target scene. The raw metric description is presented in Section 4.1; the main-flow decision is made using the controlled-augmentation requirements in Section 4.4.

3.8 Shared metric core

Both flows use a dependency-injected metric module built from NumPy, Pillow, and an injected detector callable. The module is useful only when each metric is read according to what it actually measures:

- Detection. PERSON_DETECTED and PERSON_CONFIDENCE state whether any person is detected in the output. In crowded CityPersons images this is not target-specific, because source pedestrians can keep the flag positive even when insertion fails.
- Inclusion. INCLUSION_COUNT_DELTA equals output person count minus source person count. In the supplied LoRA CSV, OBJECT_ADDED is exactly $\mathbb{1}[\Delta n > 0]$ for every row. This is a useful proxy only; it does not prove that the detector found the inserted target.
- Target-specific detection overlap. Let B_{ins} be the final inserted-object box derived from the saved insertion mask and let $\mathcal{D}$ be the person detections in the final composite. Define target_detected $= 1$ when at least one $d \in \mathcal{D}$ has confidence at least $\tau = 0.25$ and $\mathrm{IoU}(B_d, B_{\text{ins}}) \geq 0.50$. The companion values are the maximum overlapping detector confidence and $\max_{d \in \mathcal{D}} \mathrm{IoU}(B_d, B_{\text{ins}})$. These fields are specified for the paired evaluation but are not present in the retained CSVs.
- Scale. SCALE_RATIO and SCALE_ERROR require a non-zero expected-height prior. A missing prior must remain missing; it must not be encoded as a perfect score.
- Background preservation. BG_MAE and BG_SSIM are computed outside implementation-defined effective support M_{eff}. A value close to zero / one measures low change under that convention, not automatically byte-exact preservation. A separate pixel-regression assertion is required for the latter claim.

3.9 Evaluation protocol and LoRA audit

A direct benchmark is not available in the retained archive. A valid paired evaluation is future work and must run both flows on the same frozen source cases, target prompts, seeds where supported, detector version and threshold, output resolution, and effective-support-mask convention. Each final output must also save its insertion mask, insertion box, detector boxes, and detector confidences so that target-specific detection overlap can be measured. The current archive instead contains three Rule-Guided smoke summaries and one $N = 100$ LoRA shared-metric CSV. Table 6 is therefore a descriptive audit of separate exports, not a direct benchmark and not a statistical ranking.

The LoRA CSV was audited programmatically before aggregation. It contains 100 rows, all labelled `lora`, with 100 unique CASE_ID values and 100 unique seeds from 42 through 141. No values are missing in detection, inclusion, or background columns. The arithmetic identity PERSON_COUNT−SOURCE_PERSON_COUNT=INCLUSION_COUNT_DELTA holds in every row, and OBJECT_ADDED equals $\mathbb{1}[\Delta n > 0]$ in every row. In contrast, EXPECTED_HEIGHT equals 0 in all rows and both scale columns are missing in all rows. Thus, the inclusion and background statistics are usable under their stated conventions, whereas LoRA scale performance is not measurable from this export.

Table 5: Metric interpretation used in this report. Detector presence and count delta are proxies; they do not prove that the detector found the newly inserted target.

Metric	Interpretation	Direction
person_detected / person_confidence	Any person detected in output; not target-specific in multi-person scenes.	higher
object_added / inclusion_count_delta	Detector-count evidence for a new person relative to source; not target-specific.	higher
target_detected / target_box_IoU	Detection overlap with the saved inserted mask or box; required for a paired benchmark.	higher
scale_ratio / scale_error	Detected versus expected height, only when expected height is non-zero.	ratio neutral; error lower
bg_mae/ bg_ssim	Difference outside a common M_{eff} support convention.	MAE lower; SSIM higher

The project code identifies YOLOv8 [2] as the detector family (YOLOv8n for detection and YOLOv8m-seg where masks are needed), using a default confidence threshold of 0.25 and an 8 px dilation before background metrics. These settings are recorded as code defaults; run-level model and configuration hashes remain required for a fully reproducible benchmark.

4 Results and Discussion

4.1 Unpaired descriptive metric exports (not a benchmark)

The supplied shared-metric files are unpaired and are not a direct benchmark. The Rule-Guided file contains 32 rows with 31 unique case IDs and 32 unique case–seed pairs. The LoRA file contains 100 unique case IDs and 100 unique case–seed pairs. The source scenes, output provenance, candidate attempts, and effective-support masks are not matched. The values below are reported for transparency, but they are not the decision criterion for the primary method because they omit the controls that define a usable scene-aware augmentation output.

The descriptive table shows that LoRA has favourable values in its own detector-count and background columns. These values are not sufficient to make LoRA the main controlled-augmentation method, and Table 6 must not be read as a benchmark. They do not measure whether the detector overlaps the inserted person, whether the person was placed in a safe region, has an expected height, contacts the ground, respects occlusion, or passes edge-quality rejection. The next paired evaluation will report target_detected, target_box_IoU, and target confidence from the saved insertion mask or box, alongside scale, placement, and acceptance metrics. Until then, primary-flow selection is an operational readiness decision, not a claim of superior benchmark performance.

LoRA metric audit

The LoRA CSV was audited rather than reported from its mean values alone. It contains 100 rows, 100 unique cases, and 100 unique seeds. Object addition is $69/100 = 0.690$ with a 95% Wilson interval $[0.594, 0.772]$. Mean inclusion-count delta is 0.720; the split is 69 positive, 23 zero, and 8 negative rows. The 98/100 person-detection rate is not target-specific because source

Table 6: Unpaired descriptive metric exports from separate CSV files. This table is not a direct benchmark: the flows use different source cases, output provenance, and effective-support-mask conventions.

Metric	Rule-Guided ($N = 32$)	LoRA ($N = 100$)	Interpretation
Object-added rate	18/32 = 0.563	69/100 = 0.690	Different cases and detector-count proxy; not target-specific.
Mean count delta	0.469	0.720	Different cases and detector-count proxy; not target-specific.
Mean person confidence	0.894	0.851	Rule-Guided is higher, but a person detector does not identify the inserted target in crowded scenes.
Mean background MAE	19.108	0.212	Not directly comparable without the same effective support and export convention.
Mean background SSIM	0.554	0.997	Not directly comparable without the same effective support and export convention.
Scale error	Not available	Not available	No valid common shared scale-error export.

pedestrians can keep the detector positive when no new pedestrian is added.

LoRA background values are low on average under its own exported support convention: mean BG_MAE 0.2120 (SD 1.1374, median 0, P95 0.8588, maximum 10.6237) and mean BG_SSIM 0.9969 (SD 0.0177, median 1.0000, minimum 0.8329). However, 20 rows have non-zero BG_MAE, 4 exceed 1.0, and 5 have BG_SSIM below 0.99. The main limitation is not a missing pixel statistic: LoRA has no usable expected-height scale information, no target-scene placement rule, and no dedicated edge-quality rejection log. The CSV therefore describes a secondary appearance-and-paste baseline, not a complete controlled-augmentation result.

4.2 Rule-guided primary-pipeline diagnostics

Table 8 aggregates the three saved Rule-Guided SD3.5 Pipeline metric summaries. Their exact source/run mapping is recorded in the reproducibility statement: `metrics_summary.csv` maps to cityperson_nguyena, `metrics_summary(1).csv` maps to cityperson_bg_yolo, and `metrics_summary(2).csv` maps to `MOT17`. Across these smoke runs, 150 candidates were generated, 103 accepted, and 47 rejected, yielding a weighted acceptance rate of 0.687.

The accepted proportion improves from 0.60 on cityperson_nguyena to 0.72 on cityperson _bg_yolo and 0.74 on `MOT17`. The reported flow-specific scale score is 0.9676, 0.9742, and 0.9284, respectively; its aggregate 0.957 is the unweighted mean of the three run-level means, not a pooled per-candidate statistic. While the saved summaries do not contain a matched shared background / inclusion export, they demonstrate that the primary flow operationalises the controls absent from the current LoRA path: target-scene placement, expected-height scale correction, ground-contact and occlusion checks, mask and edge filtering, explicit rejection labels, and bounded retry. A candidate is saved only after these checks pass.

Table 9 shows that `mask_aspect_invalid` is the dominant rejection category (28/47, 59.6%), followed by `bad_placement` (11/47, 23.4%). The result directs future work toward segmentation and placement-policy refinement, not a relaxation of quality gates. The rejected examples are useful diagnostics rather than evidence that the filter should accept weaker samples.

Table 7: LoRA metric audit: The audit keeps uncertainty and failure counts visible and identifies the missing controls that prevent the current LoRA path from serving as the main controlled-augmentation flow.

Area	**Audited result**	**Interpretation**
Row integrity	100 rows; 100 unique cases; 100 unique seeds (42–141).	No duplicate row identity or mixed-flow label.
Object addition	69/100 = 0.690; 95% Wilson [0.594, 0.772].	Detector-count proxy, not target-specific insertion proof.
Count delta	Mean 0.720; 69/23/8 positive/zero/negative.	Negative and zero rows require visual review.
Person detection	98/100 = 0.980.	Not target-specific: source people can keep this value positive.
Person confidence	Mean 0.851, SD 0.149; detected-only mean 0.868.	Detector confidence is not placement or scale quality.
Background MAE	Mean 0.2120; max 10.6237; 80 zero rows.	Low average change under LoRA's support convention.
Background SSIM	Mean 0.9969; min 0.8329; 85 rows equal 1.0.	Strong similarity average, not a controlled-placement metric.
Scale, placement, edge gate	No valid scale rows; no target-scene policy or dedicated edge filter.	Primary limitations; moved to future work.

Table 8: Three saved source/run-specific Rule-Guided SD3.5 Pipeline smoke summaries. Aggregate acceptance is total accepted / total generated; the aggregate scale value is the unweighted mean of the three run-level means.

Source/run	Generated	Accepted	Rejected	Accept rate	Mean scale
cityperson_nguyena	50	30	20	0.60	0.9676
cityperson_bg_yolo	50	36	14	0.72	0.9742
MOT17	50	37	13	0.74	0.9284
Aggregate (three runs)	150	103	47	0.687	0.957

4.3 Blinded manual review for future confirmation

The automatic metrics do not directly measure whether an inserted pedestrian is physically plausible in its target scene. They cannot fully assess safe placement, scale and ground contact, occlusion correctness, boundary artefacts, or whether the final image is suitable for detector training. No blinded manual-review scores were supplied with the archive; therefore, no manual-review result is reported in this study.

A future review should use the 40 frozen source scenes specified in Section 3.4. For each source scene, one final Rule-Guided output and one final LoRA output should be anonymised, randomly ordered, and shown without flow names, paths, prompts, metric values, or debug overlays. At least three independent reviewers should score visual realism, scene compatibility, placement plausibility, scale and grounding, boundary quality, and usefulness for dataset augmentation on a 1–5 scale. Each reviewer should also assign a binary augmentation-ready decision.

For image i and reviewer r, the per-image score may be computed as

$$S_{i,r} = \frac{1}{6}\sum_{k=1}^{6} s_{i,r,k}, \qquad s_{i,r,k} \in \{1,2,3,4,5\}. \tag{7}$$

The future report should provide the method-level mean, augmentation-ready rate, paired source-scene win/tie/loss counts, a paired Wilcoxon test, and reviewer agreement for the binary ready label using Fleiss' κ. This review will test whether the operational advantages of Rule-Guided also produce a human-visible quality advantage.

Table 9: Aggregated rejection histogram across the three supplied Rule-Guided SD3.5 Pipeline smoke summaries ($N_{\text{rejected}} = 47$). Reason labels are preserved from the CSV fields.

Reject reason	Count	Share of rejected
`mask_aspect_invalid`	28	0.596
`bad_placement`	11	0.234
`bad_person_depth_overlap`	3	0.064
`floating_or_bad_ground`	2	0.043
`scale_unrecoverable`	2	0.043
`no_person_detected`	1	0.021
Total	47	1.000

Table 10: Task-readiness decision: Rule-Guided SD3.5 Pipeline is the main flow because it covers the controls required for safe scene-aware pedestrian augmentation.

Required capability	**Rule-Guided**	**Current LoRA**	**Main flow**
Uses target-scene context before generation	Local context crop and target scene.	Prompt-only person generation.	Rule-Guided
Placement awareness	Placement policy, depth, overlap, clearance, and safe regions.	Not implemented.	Rule-Guided
Scale and grounding	Expected-height correction, foot anchor, and rejection.	Not implemented; no valid scale metric.	Rule-Guided
Occlusion handling	Foreground/overlap checks and rejection.	No comparable target-scene check.	Rule-Guided
Edge filtering	Mask cleanup, edge / seam controls, quality gate, reject reason.	Basic blend only; no dedicated edge-quality filter.	Rule-Guided
Validation and retry	Safety gate, bounded retry, rejection histogram.	No comparable acceptance/retry log.	Rule-Guided
Retained operational evidence	103/150 accepted; 47 logged rejects; mean scale score 0.957.	100 descriptive shared-metric rows; scale unavailable.	Rule-Guided

4.4 Operational decision: Rule-Guided SD3.5 Pipeline is more production-ready under current evidence

The primary outcome of this project is not merely a person-like crop or a low background difference. It is a controlled training image: the inserted pedestrian must be plausible in the target scene, have an acceptable scale, contact the ground, respect occlusion, have a clean boundary, and be recorded as accepted only after safety checks pass. Under this task definition, Rule-Guided SD3.5 Pipeline is the more production-ready controlled-augmentation flow under current evidence.

The current LoRA path is weaker for controlled augmentation because it creates a person without target-scene context, then pastes the person into a separate scene. It has no implemented target-scene placement awareness, no expected-height scale correction, no ground-contact correction, and no dedicated edge-quality filtering or rejection stage. Basic mask erosion and feathering reduce some visible seams, but they do not replace an edge-quality gate that can reject unsafe composites. These missing capabilities are formalised as future work in Section 4.5.

> "Final operating decision: Rule-Guided SD3.5 Pipeline is the main method for controlled pedestrian augmentation. The current LoRA path is a secondary appearance-generation baseline. Rule-Guided is more production-ready for the project objective under current evidence because it uses target-scene context and implements placement awareness, scale correction, grounding and occlusion checks, edge filtering,

quality gates, rejection logging, tuning controls, and bounded retry. LoRA requires these additions before it can be considered a complete controlled-augmentation flow."

The saved Rule-Guided summaries support this decision. Across three smoke runs, the pipeline generated 150 candidates, accepted 103, and rejected 47; the weighted acceptance rate is 0.687 with a 95% Wilson interval $[0.609, 0.755]$. The mean scale score is 0.957 across the three run-level means. Rejection is informative: invalid mask shape accounts for 28/47 rejects and bad placement for 11/47. These failure labels drive the tuning functions in Table 3. A dataset-production pipeline should reject unsafe samples rather than treat every detector-visible person as a valid training example.

4.5 Future work: complete the LoRA path

The future goal is not to remove LoRA, but to turn it from a prompt-only appearance baseline into a scene-aware controlled-augmentation option. The following additions are required before that comparison can be considered fair:

1. Placement awareness. Condition the LoRA generation or paste location on a target-scene placement proposal, safe-region mask, depth estimate, and clearance checks.
2. Scale correction. Estimate an expected target height from the target scene, resize the generated person and mask together, and anchor the feet to the predicted ground contact point.
3. Grounding and occlusion checks. Reject floating persons, impossible depth order, and invalid overlap with foreground people or vehicles.
4. Dedicated edge filtering. Add boundary/halo measurements, mask-shape checks, edge-seam rejection, and retry rather than relying only on basic erosion, feathering, or colour transfer.
5. Paired evaluation. Run both flows on the same frozen source images, target prompts, detector settings, output format, and effective-support-mask convention. Save the insertion mask and box for every output, then report target-specific detection overlap, target confidence, valid scale error, placement quality, edge quality, time per accepted image, and downstream detector benefit.

4.6 Compute accounting: evidence gap

The archive supports only a qualitative implementation-cost description, not a measured efficiency comparison. The rule-guided path performs placement proposal, local img2img, segmentation, scale correction, validation, and bounded retries at inference. The LoRA path pays a one-off adapter-training cost, then uses text-to-image generation and YOLOv8-seg paste at inference; the saved adapter file is reported as 29.1 MB. Because no per-case wall-clock, retry, throughput, or peak-memory logs are retained, neither runtime nor VRAM is treated as a numerical result. A future report should measure candidate time, accepted-output time, retries, peak allocated VRAM, and energy/cost on the same hardware and seeds.

4.7 Qualitative results

Figure 5 provides illustrative saved outputs rather than a paired human study. The upper examples are Type A rule-guided accepted composites and the lower examples are Type B LoRA segment-and-paste composites. The two image types must remain separate in the archive, figures, and training-data manifests. They should be inspected for paste-seam sharpness, sticker-lighting, and local scale plausibility. They reinforce the role distinction visible in the workflow diagrams:

Figure 5: **Two output types shown separately.** The upper rows are Type A rule-guided accepted composites. The lower rows are Type B LoRA prompt-generated person composites. The panels are not matched pairs and do not quantify a cross-flow gap.

Rule-Guided SD3.5 Pipeline is the scene-aware controlled-augmentation method, whereas current LoRA is a prompt-only appearance generator followed by paste. The qualitative panels should not be used as a numerical proof; a final experiment report should replace them with enlarged matched source–candidate–mask–composite–difference panels and a blinded review.

4.8 Discussion

The available evidence separates two questions. The current LoRA export has favourable detector-count and background values under its own convention, but those values are not target-specific and cannot rank two unpaired flows. Rule-Guided is selected for the present dataset-production workflow because it implements target-scene placement, expected-height scale correction, grounding and occlusion checks, edge filtering, acceptance logging, and bounded retry. The retained 103/150 accepted-candidate result and rejection histogram show that these controls are active, but they do not establish a universal visual-quality advantage.

5 Conclusion and Perspective

This study identifies Rule-Guided SD3.5 Pipeline as the more production-ready controlled-augmentation flow under current evidence. It is selected because it provides scene-aware acceptance and rejection controls required to create controlled pedestrian training images. The current LoRA path remains a useful appearance-generation baseline, but it lacks target-scene placement awareness, valid expected-height scale measurement, and a dedicated edge-quality rejection stage.

The shared CSV values are descriptive only, not a direct benchmark. A future paired evaluation will run both flows on common frozen scenes and will report target-specific detection overlap with the saved inserted mask or box, valid scale error, placement and grounding outcomes, background change under one effective-support convention, runtime per accepted output, and blinded manual review. Until that experiment is completed, the appropriate claim is operational readiness rather than cross-flow superiority.

Reproducibility

The report source identifies the following artifacts: the LoRA configuration files under `LoRA/configs/*.yaml`; the training and apply notebooks `concept_01_all_in_one.ipynb` and `concept_02_segment_paste.ipynb`; the adapter `pytorch_lora_weights.safetensors`; per-row seeds in `lora_shared_metrics.csv` beginning at 42; the audit artifacts `lora_metrics_audit.json`, `lora_metrics_audit.csv`, `v5_shared_metrics.csv`, `rule_guided_lora_shared_metric_comparison.csv`, and `rule_guided_lora_metrics_audit.json`; and the three rule-guided smoke files `metrics_summary.csv`, `metrics_summary(1).csv`, and `metrics_summary(2).csv`. For traceability, their retained mapping is cityperson_nguyena, cityperson_bg_yolo, and `MOT17`, respectively. Research autotune is intended to remain in `dry_run` so parameters do not change silently.

For the future paired benchmark, each run must also write a machine-readable `run_manifest.json` containing the Git commit, base-model revision, adapter hash, package versions, resolved configuration hash, dataset release and split hashes, source-case identifiers, seed list, detector model and threshold, effective-support-mask path and hash, insertion mask and box, output hash, and timing/VRAM records. This manifest is required to reproduce both automatic metrics and target-specific overlap calculations.

Several items required for full regeneration are not retained in the supplied archive and are therefore not claimed: the exact Git commit, immutable resolved training command, package versions, base-model revision, complete dataset release manifest and hashes, matched baseline shared-metric CSV, common effective-support masks, target-specific matching evidence, and per-case timing/VRAM logs. A future release should store these items alongside the adapter and include SHA-256 hashes for the dataset manifest, configuration, training script, and final adapter.

The 40-source / 200-candidate protocol is future work, not observed data. This report uses the retained 150 Rule-Guided candidate summaries, 32 Rule-Guided shared-metric rows, and 100 LoRA shared-metric rows exactly as supplied. No paired target-specific evaluation or blinded manual-review result is claimed.

References

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A Layer-by-layer model specification

The intended trained component is a LoRA adapter on the attention projections of the SD3.5 MM-DiT backbone; the VAE, text encoders, and base MM-DiT weights are configured to remain frozen. Run-level provenance is needed to verify that the reported adapter used precisely this configuration. The adapter (Table 11) injects a low-rank update $\Delta W = BA$ with rank $r = 16$ into the attention `to_q/to_k/to_v/to_out` projections of each `JointTransformerBlock`; the base projection weights are untouched.

Table 11: The single trained component (the attention LoRA). Frozen modules are listed for completeness.

Component	Role	Trained?
VAE encoder/decoder	image $\leftrightarrow$ 16-ch latent	frozen
CLIP-L / CLIP-G / T5-XXL	text conditioning (T5 droppable)	frozen
MM-DiT base weights	denoiser backbone	frozen
Attention LoRA (r=16)	low-rank ΔW on `q/k/v/out`	trained

B Training configuration

The following table records the configuration stated in the report source for the concept LoRA. It should be read as a configured experiment profile, not as independently verified run provenance: the supplied archive does not include an immutable resolved command, dtype log, or configuration hash for the reported adapter. The intended setup is an attention-only concept LoRA with frozen base modules and precomputed text embeddings, optimised with the SD3 rectified-flow objective and `logit_normal` timestep weighting.

Recommended accompanying columns are `reviewer_id`, `blind_image_id`, `source_pair_id`, `review_order`, `review_timestamp`, and `comment`. The true flow label is joined only after all reviewers have submitted their scores.

Table 12: Concept-LoRA configuration stated in `LoRA/configs/concept_lora_train.yaml`; run-level provenance must verify the actual reported execution.

Setting	Value
Backbone	SD3.5 Medium (bf16 compute, fp32 trainable)
Trained params	attention LoRA, rank 16
Resolution	512×512
Learning rate	1×10^{-4}
Max train steps	4000 (checkpoint every 500)
Gradient accumulation	4 (effective batch 4)
Timestep weighting	`logit_normal`
Train samples	4000 PIPE person images